

MTH166

Lecture-4

Equations Reducible to Exact Form-II

Topic:

Equations of Reducible to Exact form

Learning Outcomes:

1. How to find I.F. using different methods (Next two methods)
2. How to convert a non-exact equation into an exact equation using I.F. and find its solution

Methods to find I.F.

3. If an equation: $Mdx + Ndy = 0$

can be re-written as:

$$f(xy)ydx + g(xy)x dy = 0$$

then, $I.F. = \frac{1}{Mx - Ny}$ provided $Mx - Ny \neq 0$

Solve the following differential equations:

Problem 1. $(y + xy^2)dx + (x - x^2y)dy = 0$ (1)

Solution: $(y + xy^2)dx + (x - x^2y)dy = 0$

Comparing it with: $Mdx + Ndy = 0$

Here $M = (y + xy^2) \Rightarrow \frac{\partial M}{\partial y} = 1 + 2xy$

And $N = (x - x^2y) \Rightarrow \frac{\partial N}{\partial x} = 1 - 2xy$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So, Equation (1) is Non-exact .

Since equation (1) can be re-written as:

$$(1 + xy)ydx + (1 - xy)x dy = 0$$

Which is of the form: $f(xy)ydx + g(xy)x dy = 0$

$$I.F. = \frac{1}{Mx - Ny} = \frac{1}{(y + xy^2)x - (x - x^2y)y} = \frac{1}{2x^2y^2}$$

Multiplying equation (1) by I.F.

$$[(y + xy^2)dx + (x - x^2y)dy] \times \frac{1}{2x^2y^2} = 0$$

$$\Rightarrow \left(\frac{1}{2x^2y} + \frac{1}{2x}\right) dx + \left(\frac{1}{2xy^2} - \frac{1}{2y}\right) dy = 0$$

Which is an exact differential equation.

$$\text{Solution: } \int_{y=\text{con.}} Mdx + \int (\text{Terms of } N \text{ free from } x) dy = C$$

$$\Rightarrow \int_{y=\text{con.}} \left(\frac{1}{2x^2y} + \frac{1}{2x}\right) dx - \int \left(\frac{1}{2y}\right) dy = C$$

$$\Rightarrow -\frac{1}{2xy} + \frac{1}{2} \log x - \frac{1}{2} \log y = c \text{ **Answer.**}$$

Solve the following differential equations:

Problem 2. $(x^2y^3 - y)dx + (x^3y^2 + x)dy = 0$ (1)

Solution: $(x^2y^3 - y)dx + (x^3y^2 + x)dy = 0$

Comparing it with: $Mdx + Ndy = 0$

Here $M = (x^2y^3 - y) \Rightarrow \frac{\partial M}{\partial y} = 3x^2y^2 - 1$

And $N = (x^3y^2 + x) \Rightarrow \frac{\partial N}{\partial x} = 3x^2y^2 + 1$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So, Equation (1) is Non-exact .

Since equation (1) can be re-written as:

$$(x^2y^2 - 1)ydx + (x^2y^2 + 1)x dy = 0$$

Which is of the form: $f(xy)ydx + g(xy)x dy = 0$

$$I.F. = \frac{1}{Mx - Ny} = \frac{1}{(x^2y^3 - y)x - (x^3y^2 + x)y} = \frac{-1}{2xy}$$

Multiplying equation (1) by I.F.

$$[(x^2y^3 - y)dx + (x^3y^2 + x)dy] \times \frac{-1}{2xy} = 0$$

$$\Rightarrow \left(-\frac{xy^2}{2} + \frac{1}{2x}\right) dx + \left(-\frac{x^2y}{2} - \frac{1}{2y}\right) dy = 0$$

Which is an exact differential equation.

$$\text{Solution: } \int_{y=\text{con.}} Mdx + \int (\text{Terms of } N \text{ free from } x) dy = C$$

$$\Rightarrow \int_{y=\text{con.}} \left(-\frac{xy^2}{2} + \frac{1}{2x}\right) dx - \int \left(\frac{1}{2y}\right) dy = C$$

$$\Rightarrow -\frac{x^2y^2}{4} + \frac{1}{2}\log x - \frac{1}{2}\log y = c \text{ **Answer.**}$$

Polling Quiz

For what values of a , given differential equ.: $(x^2 - ay)dx + (y^2 - ax)dy = 0$ is exact:

(A) For all values of a (Answer)

(B) There does not exist any value of a for which given diff. equ. is exact.

(C) Only for $a = 1$

(D) None of these.

Methods to find I.F.

4. For an equation: $Mdx + Ndy = 0$ (1)

(I) If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$, then $I.F. = e^{\int f(x)dx}$

(II) If $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = f(y)$, then $I.F. = e^{\int f(y)dy}$

Problem 1. $(4xy + 3y^2 - x)dx + (x^2 + 2xy)dy = 0$ (1)

Solution: $(4xy + 3y^2 - x)dx + (x^2 + 2xy)dy = 0$

Comparing it with: $Mdx + Ndy = 0$

Here $M = (4xy + 3y^2 - x) \Rightarrow \frac{\partial M}{\partial y} = 4x + 6y$

and $N = (x^2 + 2xy) \Rightarrow \frac{\partial N}{\partial x} = 2x + 2y$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So, Equation (1) is Non-exact .

Here $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{(4x+6y)-(2x+2y)}{(x^2+2xy)} = \frac{2(x+2y)}{x(x+2y)} = \frac{2}{x} = f(x)$

So, $I.F. = e^{\int f(x)dx} = e^{\int \frac{2}{x}dx} = e^{2 \log x} = e^{\log x^2} = x^2$

Multiplying equation (1) by I.F., we get:

$$\begin{aligned} & [(4xy + 3y^2 - x)dx + (x^2 + 2xy)dy] \times x^2 = 0 \\ \Rightarrow & (4x^3y + 3x^2y^2 - x^3)dx + (x^4 + 2x^3y)dy = 0 \end{aligned}$$

Which is an exact differential equation.

$$\text{Solution: } \int_{y=\text{con.}} Mdx + \int (\text{Terms of } N \text{ free from } x)dy = C$$

$$\Rightarrow \int_{y=\text{con.}} (4x^3y + 3x^2y^2 - x^3)dx + \int (0)dy = c$$

$$\Rightarrow \frac{4x^4y}{4} + 3 \frac{x^3}{3} y^2 - \frac{x^4}{4} = c$$

$$\Rightarrow 4x^4y + 4x^3y^2 - x^4 = 4c$$

$$\Rightarrow 4x^4y + 4x^3y^2 - x^4 = c_1 \text{ **Answer.**}$$

$$(4c = c_1)$$

Problem 2. $(xy^3 + y)dx + (2x^2y^2 + 2x + 2y^4)dy = 0$ (1)

Solution: $(xy^3 + y)dx + (2x^2y^2 + 2x + 2y^4)dy = 0$

Comparing it with: $Mdx + Ndy = 0$

Here $M = (xy^3 + y) \Rightarrow \frac{\partial M}{\partial y} = 3xy^2 + 1$

and $N = (2x^2y^2 + 2x + 2y^4) \Rightarrow \frac{\partial N}{\partial x} = 4xy^2 + 2$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So, Equation (1) is Non-exact .

Here $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{(4xy^2+2)-(3xy^2+1)}{(xy^3+y)} = \frac{(xy^2+1)}{y(xy^2+1)} = \frac{1}{y} = f(y)$

So, $I.F. = e^{\int f(y)dy} = e^{\int \frac{1}{y}dy} = e^{\log y} = y$

Multiplying equation (1) by I.F., we get:

$$\begin{aligned} & [(xy^3 + y)dx + (2x^2y^2 + 2x + 2y^4)dy] \times y = 0 \\ \Rightarrow & (xy^4 + y^2)dx + (2x^2y^3 + 2xy + 2y^5)dy \end{aligned}$$

Which is an exact differential equation.

$$\text{Solution: } \int_{y=\text{con.}} Mdx + \int (\text{Terms of } N \text{ free from } x)dy = C$$

$$\Rightarrow \int_{y=\text{con.}} (xy^4 + y^2)dx + \int (2y^5)dy = c$$

$$\Rightarrow \frac{x^2y^4}{2} + xy^2 + \frac{y^6}{3} = c$$

$$\Rightarrow 3x^2y^4 + 6xy^2 + 2y^6 = 6c$$

$$\Rightarrow 3x^2y^4 + 6xy^2 + 2y^6 = c_1 \text{ **Answer.**}$$

$$(6c = c_1)$$

Polling Quiz

Solution of differential equation: $xdy + ydx = 0$ is given by:

(A) $xy = c$ (Answer)

(B) $\frac{x}{y} = c$

(C) $x + y = c$

(D) $x - y = c$

